Week #6: Least Squares and QR Decomposition

Numerical Methods: Mathematical Foundations

June 22, 2025

Overview

Linear least squares problems arise naturally when we have more equations than unknowns, leading to overdetermined systems that generally have no exact solution. This week we develop the mathematical theory for finding the "best" approximate solution in the least squares sense, exploring both the normal equations approach and the more numerically stable QR decomposition method.

The least squares problem represents a fundamental paradigm in numerical analysis: when exact solutions don't exist, we seek optimal approximate solutions according to well-defined criteria. This leads us to explore orthogonal transformations, particularly the QR decomposition, which provides both theoretical insights and practical algorithms for solving overdetermined systems.

Our analysis emphasizes the geometric interpretation of least squares as orthogonal projection onto subspaces, the role of orthogonal matrices in preserving numerical properties, and the superior stability characteristics of QR-based methods compared to normal equations. These concepts form the foundation for understanding more advanced topics in numerical linear algebra and optimization.

1 The Linear Least Squares Problem

1.1 Problem Formulation

Definition. Linear Least Squares Problem

Given $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ and $b \in \mathbb{R}^m$, find $x \in \mathbb{R}^n$ that minimizes:

$$\|Ax - b\|_2^2 = \sum_{i=1}^{m} (a_i^T x - b_i)^2$$

where a_i^T is the i-th row of A.

The geometric interpretation reveals that we seek the point Ax in the column space of A that is closest to b in the Euclidean norm.

Theorem 1.1 (Existence and Uniqueness of Least Squares Solutions)**.** *The linear least squares problem* $\min_x \|Ax - b\|_2$ *always has a solution. The solution is unique if and only if A has full column rank.*

Proof. Existence follows from the compactness of the feasible set and continuity of the objective function. For uniqueness, if A has full column rank, then $A^T A$ is positive definite, making the normal equations have a unique solution. □

1.2 Normal Equations

The classical approach to solving least squares problems uses the normal equations.

Theorem 1.2 (Normal Equations)**.** *The vector x^* minimizes $\|Ax - b\|_2$ if and only if it satisfies the normal equations:*

$$A^T A x^* = A^T b$$

Proof. The objective function $f(x) = \|Ax - b\|_2^2$ has gradient $\nabla f(x) = 2A^T(Ax - b)$. Setting this to zero yields the normal equations. □

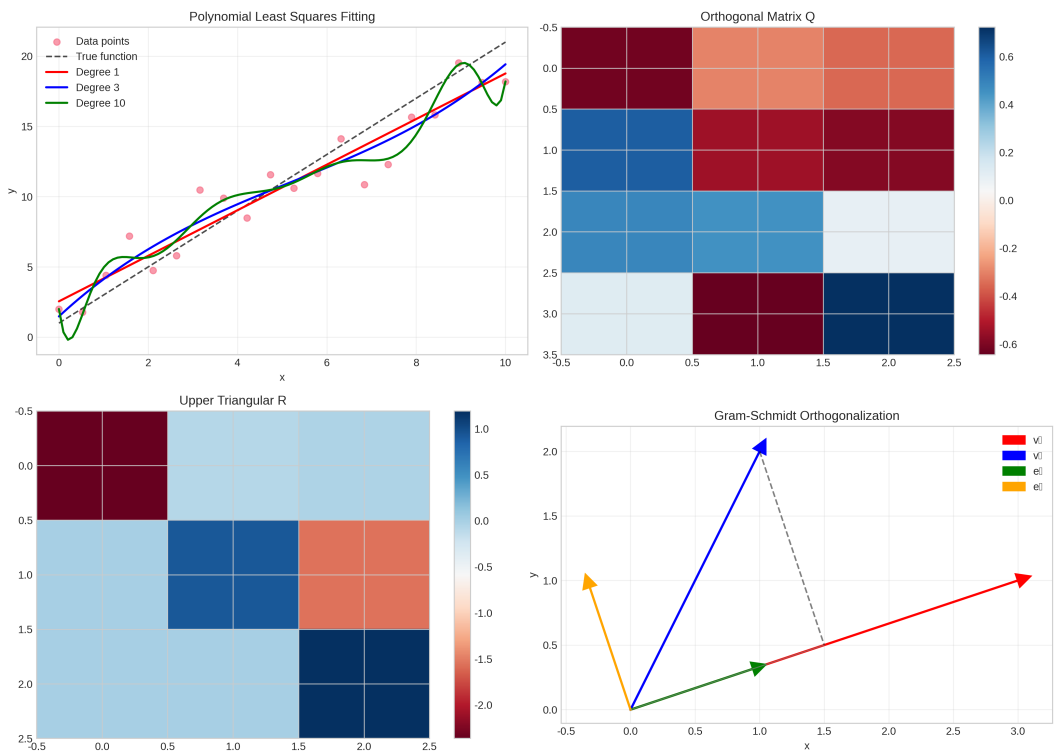

Figure 1: Geometric interpretation of least squares. The left panel shows the orthogonal projection of b onto the column space of A. The right panel illustrates the QR decomposition structure and its relationship to the least squares solution.

1.3 Geometric Interpretation

The least squares solution has a beautiful geometric interpretation in terms of orthogonal projections.

Theorem 1.3 (Orthogonal Projection Characterization)**.** *The least squares solution x^* satisfies $Ax^* = P_A b$, where $P_A = A(A^T A)^{-1} A^T$ is the orthogonal projection onto the column space of A.*

The residual $r^* = b - Ax^*$ is orthogonal to the column space of A, which gives us the fundamental relationship:

$$A^T r^* = A^T(b - Ax^*) = 0$$

2 QR Decomposition

2.1 Definition and Existence

Definition. QR Decomposition

For $A \in \mathbb{R}^{m\times n}$ with $m \geq n$, a QR decomposition is a factorization:

$$A = QR$$

where $Q \in \mathbb{R}^{m\times m}$ is orthogonal ($Q^TQ = I$) and $R \in \mathbb{R}^{m\times n}$ is upper triangular.

For computational purposes, we often use the reduced QR decomposition:

$$A = Q_1R_1$$

where $Q_1 \in \mathbb{R}^{m\times n}$ has orthonormal columns and $R_1 \in \mathbb{R}^{n\times n}$ is upper triangular.

Theorem 2.1 (Existence of QR Decomposition)**.** *Every matrix $A \in \mathbb{R}^{m\times n}$ has a QR decomposition. If A has full column rank, then R can be chosen to have positive diagonal entries, making the decomposition unique.*

2.2 Gram-Schmidt Process

The classical Gram-Schmidt process provides a constructive proof of QR decomposition existence.

Definition. Classical Gram-Schmidt

Given linearly independent vectors $a_1, \ldots, a_n$, the Gram-Schmidt process constructs orthonormal vectors $q_1, \ldots, q_n$:

$$v_1 = a_1, \quad q_1 = \frac{v_1}{\|v_1\|} \tag{1}$$

$$v_k = a_k - \sum_{j=1}^{k-1}(q_j^T a_k)q_j, \quad q_k = \frac{v_k}{\|v_k\|} \tag{2}$$

This process is numerically unstable due to loss of orthogonality in finite precision arithmetic.

Definition. Modified Gram-Schmidt

The modified Gram-Schmidt process improves numerical stability:

$$v_1^{(1)} = a_1, \quad q_1 = \frac{v_1^{(1)}}{\|v_1^{(1)}\|} \tag{3}$$

$$v_k^{(k)} = a_k \tag{4}$$

$$v_k^{(j)} = v_k^{(j-1)} - (q_{j-1}^T v_k^{(j-1)})q_{j-1}, \quad j = 2, \ldots, k \tag{5}$$

$$q_k = \frac{v_k^{(k)}}{\|v_k^{(k)}\|} \tag{6}$$

2.3 Householder Reflections

Householder reflections provide the most numerically stable method for computing QR decompositions.

Definition. Householder Reflection

For a nonzero vector $v \in \mathbb{R}^m$, the Householder reflection is:

$$H = I - 2\frac{vv^T}{\|v\|^2}$$

This matrix is symmetric, orthogonal, and reflects vectors across the hyperplane orthogonal to v.

Theorem 2.2 (Householder QR Algorithm)**.** *For any $A \in \mathbb{R}^{m\times n}$, there exist Householder matrices $H_1, \ldots, H_n$ such that:*

$$H_n \cdots H_1 A = R$$

where R is upper triangular. The QR decomposition is $A = QR$ with $Q = H_1 \cdots H_n$.

The Householder vector for eliminating entries below the diagonal in column k is chosen as:

$$v_k = x_k + \text{sign}(x_{k,k})\|x_k\|e_1$$

where x_k is the k-th column of the current matrix.

3 Solving Least Squares via QR

3.1 QR-Based Solution

Using the QR decomposition, the least squares problem becomes much simpler to solve.

Theorem 3.1 (Least Squares Solution via QR)**.** *If $A = QR$ where Q is orthogonal and R is upper triangular, then the least squares solution is:*

$$x^* = R^{-1}Q^T b$$

assuming R is invertible (i.e., A has full column rank).

Proof. Since Q is orthogonal, $\|Ax - b\|_2 = \|QRx - b\|_2 = \|Rx - Q^T b\|_2$. The minimum occurs when $Rx^* = Q^T b$. □

For the reduced QR decomposition $A = Q_1 R_1$:

$$x^* = R_1^{-1} Q_1^T b$$

3.2 Computational Algorithm

The complete algorithm for solving least squares via QR decomposition:

1. Compute QR decomposition: $A = QR$ (using Householder reflections)
2. Compute $c = Q^T b$
3. Solve upper triangular system: $Rx = c$

The computational cost is approximately $2mn^2 - \frac{2n^3}{3}$ operations for the QR decomposition, plus $O(n^2)$ for the triangular solve.

4 Numerical Stability Analysis

4.1 Condition Numbers

The conditioning of least squares problems depends on the matrix A.

Definition. Condition Number for Least Squares

For the least squares problem $\min_x \|Ax - b\|_2$, the condition number is:

$$\kappa(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

where $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of A.

Theorem 4.1 (Perturbation Analysis)**.** *For small perturbations ΔA and Δb, the relative error in the least squares solution satisfies:*

$$\frac{\|\Delta x\|}{\|x\|} \leq \kappa(A)\left(\frac{\|\Delta A\|}{\|A\|} + \frac{\|\Delta b\|}{\|b\|}\right) + O(\epsilon^2)$$

4.2 Stability Comparison

Theorem 4.2 (Stability of Normal Equations vs. QR)**.**

- *Normal equations: The computed solution satisfies $(A^TA + \Delta(A^TA))\tilde{x} = A^Tb$ where $\|\Delta(A^TA)\| \leq \gamma_n\|A^TA\|$ with $\gamma_n = O(n\epsilon_{mach})$.*
- *QR decomposition: The computed solution satisfies $(A + \Delta A)\tilde{x} = b$ where $\|\Delta A\| \leq \gamma_n\|A\|$.*

The key insight is that the condition number of A^TA is the square of the condition number of A, making normal equations potentially much less stable.

Intuition: Why QR is More Stable

The normal equations square the condition number because we compute A^TA. QR decomposition works directly with A, preserving its conditioning. Orthogonal transformations (like those in QR) preserve norms and angles, leading to better numerical behavior.

5 Rank-Deficient Problems

When A does not have full column rank, the least squares problem has infinitely many solutions.

Definition. Minimum Norm Solution

Among all least squares solutions, the minimum norm solution x^+ is the unique solution with smallest Euclidean norm:

$$x^+ = \arg\min\{\|x\|_2 : \|Ax - b\|_2 = \min_y \|Ay - b\|_2\}$$

Theorem 5.1 (Minimum Norm Solution via Pseudoinverse)**.** *The minimum norm solution is given by:*

$$x^+ = A^+b$$

where $A^+ = (A^TA)^+A^T$ is the Moore-Penrose pseudoinverse.

For rank-deficient problems, we can use:

- QR decomposition with column pivoting
- Singular Value Decomposition (SVD)
- Regularization techniques

6 Applications and Extensions

6.1 Weighted Least Squares

When observations have different reliabilities, we use weighted least squares.

Definition. Weighted Least Squares

Given weights $w_1, \ldots, w_m > 0$, minimize:

$$\sum_{i=1}^{m} w_i (a_i^T x - b_i)^2 = \|W^{1/2}(Ax - b)\|_2^2$$

where $W = \mathrm{diag}(w_1, \ldots, w_m)$.

This reduces to the standard least squares problem $\min_x \|\tilde{A}x - \tilde{b}\|_2$ where $\tilde{A} = W^{1/2}A$ and $\tilde{b} = W^{1/2}b$.

6.2 Recursive Least Squares

For updating solutions when new data arrives:

Theorem 6.1 (Sherman-Morrison-Woodbury Formula Application)**.** *If we have solved $A_k^T A_k x_k = A_k^T b_k$ and receive new data (a_{k+1}, b_{k+1}), the updated solution can be computed in $O(n^2)$ operations using:*

$$x_{k+1} = x_k + \frac{P_k a_{k+1}(b_{k+1} - a_{k+1}^T x_k)}{1 + a_{k+1}^T P_k a_{k+1}}$$

where $P_k = (A_k^T A_k)^{-1}$ and P_{k+1} is updated accordingly.

Practice Problems

1. Prove that the normal equations $A^T Ax = A^T b$ are equivalent to the condition that the residual $r = b - Ax$ is orthogonal to the column space of A.
2. Show that the condition number of $A^T A$ equals $\kappa(A)^2$ where $\kappa(A)$ is the condition number of A.
3. Implement both classical and modified Gram-Schmidt and compare their numerical behavior on increasingly ill-conditioned matrices.
4. Derive the Householder vector formula for reflecting a vector x to $\pm\|x\|e_1$.
5. Analyze the computational complexity of solving least squares problems using: (a) Normal equations, (b) QR decomposition, (c) SVD.

Solution:

Problem 1: The normal equations state $A^T Ax = A^T b$, which can be rewritten as $A^T(Ax - b) = 0$. This means $Ax - b$ is orthogonal to every column of A, hence orthogonal to the column space of A.

Problem 2: Using the SVD $A = U\Sigma V^T$, we have $A^T A = V\Sigma^2 V^T$. The condition number of $A^T A$ is $\sigma_{\max}^2/\sigma_{\min}^2 = (\sigma_{\max}/\sigma_{\min})^2 = \kappa(A)^2$.

Problem 3: Modified Gram-Schmidt maintains better orthogonality because it subtracts projections immediately rather than accumulating them. For ill-conditioned matrices, classical Gram-Schmidt can lose orthogonality completely.

Problem 4: To reflect x to $\pm\|x\|e_1$, choose $v = x \mp \|x\|e_1$ (with sign chosen to avoid cancellation). Then $H = I - 2vv^T/\|v\|^2$ gives $Hx = \pm\|x\|e_1$.

Problem 5: Complexities are: (a) Normal equations: $mn^2 + n^3/3$, (b) QR: $2mn^2 - 2n^3/3$, (c) SVD: $2mn^2 + 11n^3$. QR is typically preferred for overdetermined systems.